LIN JIE

Phone: (+65) 9395-6657 | (+86) 181-7628-3136 | Email: Lam_Git_2002@163.com
LinkedIn: linkedin.com/in/jie-lin023 | Portfolio: afay-023.github.io/

EDUCATION

Nanyang Technological University (NTU) | Data Science | Master's **Singapore, Jan 2026 – Present**

Nanjing University | Urban & Rural Planning | Bachelor's **Nanjing, Sep 2020 – Jun 2024**

GPA: 4.31/5.0 | Average Score: 86.2/100

Honors: People's Scholarship First Prize (2022); Smart City Scholarship (2023); NJU Outstanding Undergraduate Thesis First Prize (2024)

WORK EXPERIENCE

Nightingale Health — AI Innovation Incubation | AI Product Intern (Industry Mentorship) **Singapore, Jan 2026 – Present**

Joined an internal venture team to help build a clinical AI Copilot for GP clinic workflows from 0 to 1

- **Scenario Definition & Demo Validation:** Defined two core scenarios — real-time escalation of high-risk conversations to clinics and automated ongoing care guidance based on physician orders; independently authored PRDs and built runnable demos to drive early-stage alignment and solution discussion.
- **Long-term Memory Design & MVP Architecture:** Designed a patient long-term memory pipeline (conversation → LLM extraction → structured JSON → vector retrieval) and aligned the medical ontology with FHIR; in architecture review, decomposed needs into semantic retrieval, structured entity relations, and graph reasoning, and selected Supabase + PGVector as the initial architecture based on business priorities and implementation boundaries.
- **Retrieval Evaluation & Product Decision Support:** For the patient long-term memory scenario, decomposed retrieval into within-patient longitudinal symptom/diagnosis retrieval and cross-patient similar-case retrieval; built a multilingual patient-aware benchmark, compared 3 Azure Embedding models, and cross-validated results with external benchmarks to support V1 retrieval selection.
- **Privacy Compliance & De-identification:** Built a PHI de-identification benchmark to compare Regex, Presidio NER, and LLM approaches on privacy recall and semantic fidelity; recommended Azure Language Text PII Redaction as the production solution to meet PDPA compliance requirements.

Nanjing University, School of Architecture & Urban Planning | Research Assistant **Nanjing, Sep 2024 – Jun 2025**

Conducted quantitative analysis on the spatial network structure and regional resource allocation of the Nanjing Metropolitan Area

- **Multi-source Data Integration:** Integrated 5+ heterogeneous data sources including mobile signaling and POI data using Python, building a multi-dimensional urban database covering 60 districts/counties (500K+ records) with a reusable data collection and cleansing pipeline.
- **Quantitative Analysis & Decision Support:** Applied gravity models to quantify regional spatial interaction intensity, validated clustering patterns using Moran's I spatial autocorrelation, and produced 40+ thematic analysis maps with resource allocation optimization recommendations to support project reporting and policy communication.

PROJECTS

FreshSite Copilot | Fresh Food Siting Decision Copilot **Jan-Jun 2024 (Prototype) | Mar 2026 (Demo)**

Productized a knowledge-graph siting research prototype into a public-facing explainable decision copilot demo

- **Problem Discovery & Goal Definition:** Through surveys and fieldwork, identified online fresh-retail competition as a key factor overlooked by traditional siting analysis, and framed the decision-support goal as candidate recommendation plus rationale explanation.
- **Heterogeneous Knowledge Graph Construction:** Built a Neo4j knowledge graph covering people, space, and online service relations (7,000+ nodes, 20,000+ edges); used Pandas for multi-source data cleaning, ETL, and entity alignment to support candidate analysis and explanation.
- **Candidate Identification & Productized Demo:** Combined GDS graph algorithms, link prediction, and rule-based reasoning to identify 13+ candidate optimization sites; rebuilt the research prototype into a public copilot demo with goal input, candidate recommendation, map interaction, and rationale explanation.

Industrial Development Strategic Plan for Cao County E-Commerce Zone | Project Member **May 2023**

- **Data Infrastructure & BI Dashboard:** Built a Power BI dashboard integrating 5 types of heterogeneous data including daily task logs and billing records; designed automated ETL pipelines to unify 100+ network metrics; developed brand operations and shipping-billing reconciliation models to improve key metric visibility and decision-making efficiency.
- **Regional Planning Index Model:** Performed time-series forecasting and outlet profiling based on 2022–2023 shipping volume data; designed a weighted regional potential index model incorporating growth, service gap, and scale dimensions; delivered regional planning segmentation and construction priority rankings to provide quantitative support for industrial park planning.

SKILLS

AI Product & Evaluation: RAG architecture, agent design, skill packaging, MCP tool integration & workflow orchestration, benchmark design, embedding evaluation, prompt design, and AI prototype validation; proficient with AI coding tools including Codex and Claude Code.

Programming & Data: Python, SQL, Neo4j, PostgreSQL/PGVector, Supabase.

Prototyping & Analysis: Figma, Power BI, XMind, Adobe (PS, AI).

Languages: Fluent in English as a working language (TOEFL 101).